

Capstone Project – Hotel Analytics Report

Overview

This final project builds a complete data pipeline for the Hotel Operations and Guest Experience Analytics System, a synthetic dataset covering four (4) room tiers, sixty (60) guest profiles, six hundred (600) reservations, and six hundred (600) revenue records spanning from January 2024 through June 2026. The work covered four (4) parts – building a relational MySQL database, writing analytical SQL queries, exposing the data through a FastAPI RestAPI, and building an interactive Power BI dashboard. I’ve also included an optional bonus machine learning component predicting reservation cancellations.

Database and SQL Queries

The database (HotelOperationsDB) contains four (4) related tables – RoomTiers, GuestProfiles, Reservations, and RevenueLog which are connected through GuestID, TierID, and ReservationID foreign keys. Eight (8) analytical SQL queries were written covering reservation volume, revenue, capacity, and guest behaviour. One (1) data quality note worth flagging – several guests in GuestProfiles share identical first and last names (ex. two (2) guests both named “David Chen” and two (2) named “Michael Iqbal”). Any analysis grouping by name alone rather than GuestID would silently merge these distinct guests together so all guest level queries and dashboard visuals in this project group by GuestID first.

Business Insights – Quarterly Revenue Trend

Quarter	Total Revenue (\$)
2024 Q1	28,608.41
2024 Q2	44,711.26
2024 Q3	33,258.69
2024 Q4	22,058.13
2025 Q1	27,240.24
2025 Q2	33,280.16
2025 Q3	34,883.13
2025 Q4	39,917.17
2026 Q1	28,044.23
2026 Q2	34,896.07

Revenue fluctuates quarter to quarter with no clear long term upward or downward trend. The strongest quarter was 2024 Q2 (\$44,711.26) and the weakest quarter was 2024 Q4 (\$22,058.13), which is a difference of roughly 103% between the best and the worst. 2025 shows a steadier, gradually climbing pattern, while 2026 opens lower before recovering in Q2. Overall this leads to stable, seasonally variable business rather than one experiencing sustained growth or decline.

Business Insights – Quarterly Revenue vs. the \$50,000 Target

Every quarter fell short of the \$50,000 benchmark:

- Closest to target was the 2024 Q2 quarter at \$44,711.26 (which was short of - \$5,288.74 or 89.4% of the target).
- Furthest from the target was the 2024 Q4 quarter at \$22,058.13 (which was short of -\$27,941.87 or 44.1% of the target).
- The average shortfall across all ten (10) quarters was about \$17,500 (which was about 65% of the target achieved on average).

No quarter reached \$50,000. This is a useful stress test of the target itself since actual revenue never exceeds \$45,000 in any observed quarter, a \$50,000 target appears set above the property’s demonstrated ceiling under current conditions. This \$50,000 figure was used as a placeholder throughout the project (SQL Query 4, the /quarterly-summary/ endpoint, and the dashboard’s Total Quarterly Revenue KPI), not derived from the hotel’s actual performance or stated business goal. A more data grounded target might instead be set relative to the trailing average of around \$32,700 a quarter or the best historical quarter.

Business Insights – Revenue by Room Tier vs. Reservation Volume by Room Tier

Room Tier	Reservations (Share)	Total Revenue (\$)
Economy	266 (44.3%)	≈ 76,000
Premium	168 (28.0%)	≈ 78,000
Suite	126 (21.0%)	≈ 110,000
Penthouse	40 (6.7%)	≈ 67,000

Suites generate the most revenue (around \$110,000) despite ranking third in booking volume which is well behind Economy’s 266 bookings. This is driven by Suite’s higher nightly rate (\$250 base vs. Economy’s \$95) more than offsetting its lower booking count. Penthouse has both the fewest bookings and the lowest revenue, suggesting possible underutilization relative to its premium positioning at its \$450 base rate which could be a potential area for targeted marketing or repricing.

Business Insights – Occupancy Rate Trend Across the Year

Monthly occupancy rate (room nights booked ÷ room nights available) fluctuates between roughly 0.5% and 2.6% month to month with two (2) peaks – April 2024 and August 2025 (both ~2.6%) and one sharp dip in October 2024 (~0.5%, which is the lowest point in the dataset). Aside from that October 2024 dip, there’s no strong repeating seasonal pattern year over year although peaks and troughs occur at different months in 2024 vs. 2025, suggesting occupancy is driven more by booking level variations than a predictable seasonal cycle.

Business Insights – Top 5 Guests by Lifetime Spend

Guest	Nights Stayed	Lifetime Spend
Aiden Khan	62	12,457.77
Bilal Thomas	40	11,415.02
Hamza Chen	44	11,374.76
Oliver Moore	48	10,490.48
Amelia Iqbal	44	10,200.26

Aiden Khan is the clear top guest in both nights stayed (62 nights which is roughly 50% more than the next highest guest) and lifetime spend. The remaining four (4) are tightly clustered between \$10,200 and \$11,415. This top five (5) group collectively accounts for \$55,938.29 or about 17% of total revenue from just five (5) of sixty (60) guest profiles, indicating a modest but present concentration of revenue amount repeat guests.

Business Insights – Room Tier Capacity Status

Room Tier	Nights Booked	Occupancy Rate (%)	Status
Economy	659	1.82	Within Capacity
Premium	411	1.62	Within Capacity
Suite	354	2.45	Within Capacity
Penthouse	121	2.23	Within Capacity

Against a 70% capacity threshold, no room tier is close to exceeding capacity with all four (4) sitting under a 2.5% occupancy when measured across the full room nights available in the dataset window. This low percentage reflects the measurement basis (600 reservations spread across ~900 days and 90 rooms total) rather than genuine under booking; any single day only has a handful of active stays out of full inventory.

REST API

The FastAPI application exposes all seven (7) required endpoints (reservations, room tiers, guest profiles, revenue log, plus three (3) analytical endpoints – reservations by tier, quarterly summary, and capacity alerts), backed directly by the MySQL database via SQLAlchemy. These endpoints were left open/unauthenticated since the proposal's Task 3 specification listed only plain GET routes with no authentication requirement. All seven (7) endpoints were verified individually in Swagger UI, each returning a 200 OK response with data matching the corresponding SQL query results exactly which confirms the consistency between the raw SQL analysis and the live API layer.

Design Rationale

The Power BI dashboard uses a clean layout built around three (3) KPI cards (Total Quarterly Revenue, Active Reservations, and Average Length of Stay), a Quarter/Year slicer that filters the whole page, and five (5) supporting visuals (bar chart, line chart, donut chart, and two (2) tables) covering revenue, occupancy, guests, and capacity questions. All chronological fields (quarter labels, month labels) use a dedicated numeric sort key column and the Sort by Column property to guarantee true chronological ordering rather than default alphabetical sorting.

Bonus – Cancellation Prediction using Machine Learning

As an optional extension beyond the four (4) required tasks, a binary classification model was built to predict whether a reservation will be cancelled, using the ML relevant fields suggested in the dataset's README (LoyaltyLevel, AgeGroup, BookingSource, NumberOfGuests, BookingLeadDays, StayLengthNights, and TierName). TotalRevenue was deliberately excluded as a feature since it is only known after a stay is completed and would leak information not available at the time of prediction.

Two (2) classifiers were trained on an 80/20 train test split and was compared:

Model	Accuracy (%)	Recall
Decision Tree	77.5	0.13
KNN	82.5	0.13

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Decision Tree Accuracy: 0.7750
      precision    recall  f1-score   support

     0       0.82      0.93      0.87        97
     1       0.30      0.13      0.18         23

   accuracy          0.78        120
  macro avg          0.56      0.53      0.53        120
weighted avg          0.72      0.78      0.74        120

Confusion Matrix (Decision Tree):
[[90  7]
 [20  3]]
=====
KNN Accuracy: 0.8250
      precision    recall  f1-score   support

     0       0.83      0.99      0.90        97
     1       0.75      0.13      0.22         23

   accuracy          0.82        120
  macro avg          0.79      0.56      0.56        120
weighted avg          0.81      0.82      0.77        120

Confusion Matrix (KNN):
[[96  1]
 [20  3]]
=====
Decision Tree Feature Importances:
LoyaltyLevel: 0.0550
AgeGroup: 0.0708
BookingSource: 0.1004
NumberOfGuests: 0.1170
BookingLeadDays: 0.4159
StayLengthNights: 0.1759
TierName: 0.0651
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At first glance KNN's higher accuracy looks like the better model, but this is misleading given the dataset's class imbalance. Roughly 81% of reservations in the test set were not cancelled, so a model that predicted "not cancelled" for almost every case would already score close to 80% accuracy without learning anything meaningful. Both models caught only 3 of 23 actual cancellations in the test set (with a 13% recall), meaning neither reliably identifies the reservations that matter most for this business. Accuracy alone is not a trustworthy metric here.

The Decision Tree's features offer a more useful takeaway. BookingLeadDays was by far the strongest predictor of cancellation (41.6% importance), followed by StayLengthNights (17.6%) and NumberOfGuests (11.7%). This aligns with the dataset's documented pattern that "long lead time bookings are somewhat more likely to cancel." A production ready model would likely need class rebalancing (ex. oversampling cancelled reservations or applying class weights) to meaningfully improve recall on the minority class.

Conclusion

The finished solution turns the Hotel Operations dataset's raw reservation and revenue data into four (4) usable outputs – a relational database, a secured scope FastAPI for programmatic access, an interactive Power BI dashboard for non-technical stakeholders, and a bonus predictive model for cancellation risk. The clearest business takeaways are:

1. Suites generate outsized revenue relative to its booking volume, suggesting room tier pricing is a stronger revenue lever than occupancy volume alone;
2. No room tier is near capacity under current booking patterns;
3. The illustrative \$50,000 quarterly target exceeds every quarter of actual historic performance, indicating it should be recalibrated against real data; and
4. Booking lead time is the strongest available signal for cancellation risk, though a naïve classifier alone struggles to translate that signal into reliable predictions given the class imbalance in the data.